Ryan Tsai

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EDUCATION

BE (Hons) Software Engineering, University of Auckland; 2024 – 2027

- *Grade*: 8.18/9.00 GPA cumulative. 8.50/9.00 within specialisation
- *Awards*: Dean's List 2025 (Top 5%)
- *Relevant Modules*: Intro to Computer Systems; Object Oriented Programming; Discrete Mathematics; Data Structures and Algorithms; Software Quality Assurance, Database Systems, Networks and Security, Web Programming, Software Architecture

Saint Kentigern College, Auckland; 2019 – 2023

- Proxime Accessit
- Top in School: Mathematics, English, Chemistry, Japanese

SKILLS

Languages: TypeScript, JavaScript, Java, Python, C, R, MATLAB
Frameworks & Libraries: Express.js, Node.js, React, Tailwind CSS, QMK
Tools and Skills: Git, GitHub Flow, Maven, JUnit, MongoDB, Firebase, Cloudflare R2, PITest, AutoCAD Inventor, KiCad

EXPERIENCE

Thinking2Morrow, English and Mathematics Tutor **Feb 2024 - Present**

- Tutor students from Year 1 to 13 across NCEA, IGCSE, Cambridge and IB curricula.
- Delivered personalized lessons and feedback, both in individual and group settings (up to 12 students)
- Developed key professional skills: time management, communication, collaborative planning

PROJECTS

Ferrous Fellow
Created a toy robot using **TensorFlow Lite** and **OpenCV** for facial and speech recognition. Integrated with the **OpenAI API** to analyse emotional cues and recommend music accordingly.
Tools: Python, AI/ML, Speech Recognition

MeowPad
Designed a hot-swappable number pad from scratch using **KiCAD** for PCB design, **AutoCAD Inventor** for case and plate design, and QMK library for firmware. Powered by an **Arduino Pro Micro** footprint MCU and equipped with an OLED display and EC11 rotary encoder knob. The MeowPad supports a 3D printed case with PCB mounted standoffs, as well as a stacked acrylic plate mount.
Tools: KiCad, AutoCAD, C, Arduino

RFID Record Player
Designed and built a 3D printed functional imitation of a record player powered by an **ESP32** and local **Python** serial monitor. When a "record" is placed onto the turntable, an internal RFID reader picks up on the RFID tag on the disc, which maps to the specific song's Spotify track ID. When the tonearm is in the active position, the song is played on the user's Spotify, and the record spins. When the tonearm is moved away, the record stops spinning, and the music is paused on the user's device.
Tools: Python, C, ESP32

VALORANT API Wrapper
Engineered a TypeScript wrapper using **Express.js** to access internal VALORANT API endpoints, enabling external apps to retrieve in-game data normally locked behind the game client.
Tools: TypeScript, Node.js, Express, REST APIs

Astral Bot / Astral Music
Created a modular Discord bot for server moderation and entertainment, integrating **MongoDB**, Reddit, and VALORANT API. Built advanced music features including queueing, volume control, and voice filters, later merging multiple bots into a unified system using Discord.js and **Node.js**.
Tools: Node.js, Discord.js, MongoDB